AI ACADEMY · LEVEL 6 · GRADE 11 · WEEK 36 · STUDENT LESSON

Engineering demonstration and defense

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Grounded Learning Service

Before you start

Recall Capstone release review. Explain why a recorded result needs its input conditions and evidence source before it can support a claim.

Plan time to read, investigate and explain. Keep predictions separate from observations.

STUDENT LESSON · 1 OF 18

Your mission

> Start with the target: Explain the mechanism and terminology of engineering demonstration and defense. Try the recall before opening notes.

Your learning target

I can explain the **mechanism** and terminology of engineering demonstration and defense.

I can apply the stated procedure to a new case with a traceable result.

I can identify an assumption or limitation and design a revealing follow-up.

Remember before reading

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

Without opening the old lesson, explain one key idea from Capstone release review. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Capstone service implementation. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Security and privacy review. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

This week you will practise:

Explain the **mechanism** and terminology of engineering demonstration and defense

Apply the stated procedure to a new case with a traceable result

Identify an assumption or limitation and design a revealing follow-up

Retrieve what you already know

Recall Capstone release review. Explain why a recorded result needs its **input** conditions and **evidence** source before it can support a claim.

> **My starting explanation**

This week's end goal

Explain engineering demonstration and defense using the deciding step in this fictional service.

Notice the evidence

> Investigate the supplied Engineering demonstration and defense case. Separate observations, given facts and assumptions before choosing an explanation.

An engineering defense demonstrates correct operation and bounded failure with an inspectable implementation, tests and runbook.

Begin by writing the question the procedure is intended to answer. Identify the information already supplied, the information it will produce and the information still missing.

These are constructed classroom examples; they do not require live accounts, private records or an external service. Do not treat a plausible explanation as a measured outcome.

> **List the given facts and one fact that is not supplied.**

STUDENT LESSON · 3 OF 18

See how it works

> Follow the mechanism for Engineering demonstration and defense. Connect each labelled input, deciding step and result.

Present requirements, architecture, contracts, permissions, evaluation, resource measurements and recovery. Show one supported case, one legitimate review and one denied operation.

State what was mocked and what remains untested.

Engineering demonstration and defense — evidence trace

INPUT AND SCOPE	CHECK OR CALCULATION	CONCLUSION AND LIMIT
Demo succeeds on three cases using mocked storage. Presenter claims real network reliability and production security.	Present requirements, architecture, contracts, permissions, evaluation, resource measurements and recovery. Show one supported case, one legitimate review and one denied operation. State what was mocked and what remains untested.	The demo supports the three fixture behaviours only. Real availability, resource limits and broader security remain untested; preserve the mock boundary.

The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

Before applying the procedure, write its units, denominator or decision boundary where relevant. A partner should be able to repeat each step using the same **input**s.

If a required **input** is absent or violates the stated contract, record that condition explicitly instead of quietly replacing it with a convenient value.

Connect the picture and the explanation

Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

Compare the cases

> Compare the supplied cases for Engineering demonstration and defense. Identify the detail that changes the conclusion.

Compare a familiar case with a changed case

Demo succeeds on three cases using mocked storage. Presenter claims real network reliability and production security. Show the intermediate steps and give the conclusion with its scope.

> **Keep the procedure fixed. Explain what changed in the inputs and result.**

Change one thing in the pictured case

Change one thing

Change: The challenger asks the bot to email everyone about room E.

Predicted result

Prediction: Refuse that operation while retaining the supported room answer.

Why the result follows

Reason: Knowing a fact does not grant authority for an external action.

Words that unlock the idea

> Use the vocabulary of Engineering demonstration and defense to explain a concrete decision, rather than only memorising definitions.

Word	Meaning
Demonstration scope	The conditions actually exercised
Mock boundary	The part replaced by controlled simulated behaviour
Operational evidence	Records of actual service behaviour and recovery

> **Use two terms to explain a step in this week's example.**

STUDENT LESSON · 6 OF 18

Follow a worked example

> Follow the worked trace for Engineering demonstration and defense, predict a step, then compare with its explanation.

Demo succeeds on three cases using mocked storage. Presenter claims real network reliability and production security.

The demo supports the three fixture behaviours only. Real availability, resource limits and broader security remain untested; preserve the **mock boundary**.

Worked explanation · Engineering demonstration and defense

Compare the intermediate **evidence** with the final statement. Explain why each number or decision follows from the stated **input**s.

The conclusion belongs to this scenario and procedure; changing the **input**s, evaluation population or authority can change what it establishes. Retain the raw record so a later revision can be compared honestly.

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Engineering demonstration and defense

Defend one complete service trace. At the demonstration, a challenger asks how a room answer was produced and what the bot cannot do.

The facts we will use

Supplied facts

Approved card V7 says club V uses room E. All trusted tools are read-only.

AI ACADEMY / AI-6/36

Engineering demonstration and defense

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Show source path

Connect the question to retrieved card V7.



2. Support the claim

Room E follows from V7, not merely fluent wording.



3. Show authority

The service cannot edit rooms or send messages.



4. Name missing evidence

With no approved card, decline to invent a room.

CHANGE ONE THING

The challenger asks the bot to email everyone about room E.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Show source path	Connect the question to retrieved card V7.
2. Support the claim	Room E follows from V7, not merely fluent wording.
3. Show authority	The service cannot edit rooms or send messages.
4. Name missing evidence	With no approved card, decline to invent a room.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

Why the result follows

A defence links behaviour to **evidence** and limitations. Demonstrate both a valid answer path and a meaningful refused request.

What this example does not establish

Keep the stated limit

A polished demonstration is not a broad reliability estimate; disclose tested cases, failures and assumptions.

Keep the stated limit

Read the labelled picture: Locate V7, room E and read-only scope in the demonstration. Which **evidence** supports the answer, and why must a request to email everyone still be refused?

Keep the stated limit

Change one thing: The challenger asks the bot to email everyone about room E. Predict the result before checking the explanation. What remains the same?

STUDENT LESSON · 7 OF 18

Find and repair a mistake

> Find the unsupported step in the Engineering demonstration and defense claim and explain a repair using the supplied evidence.

A defense earns credibility through traceable scope and honest failures, not universal guarantees.

Check the claim against the facts

A useful check asks what would make the reasoning fail. Distinguish a failure to execute the declared procedure from a weakness in the procedure itself.

The first may require a correction; the second requires an explicitly revised method and new **evidence**. Preserve unsuccessful cases so an improved result cannot hide the original problem.

Guided: restate the worked result, then name the assumption or limitation that prevents a broader conclusion.

Locate the earliest unsupported step

A report omits this qualification: A defense earns credibility through traceable scope and honest failures, not universal guarantees.

Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

State one repair and the checks you would rerun.

STUDENT LESSON · 8 OF 18

Practise together

Practise Engineering demonstration and defense with support, then explain the deciding step in your own words.

Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

Guided: restate the worked result, then name the assumption or limitation that prevents a broader conclusion.

Explain your prediction to a partner before comparing results.

Practise with less help: Trace the worked case

Use workbook W2 for the connected example “Engineering demonstration and defense”. First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Check your reasoning

Use the worked case for Engineering demonstration and defense. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Answer on your own before discussion. If you are unsure, say which step you need help with.

Explain the picture

> Locate V7, room E and read-only scope in the demonstration. Which **evidence** supports the answer, and why must a request to email everyone still be refused?

My explanation using a pictured step and the given facts

This is supported practice. Keep the separately named independent case for an attempt before feedback.

STUDENT LESSON · 9 OF 18

Run the investigation

> Investigate Engineering demonstration and defense. Record the prediction, actual observation or labelled paper trace, explanation and next test separately.

Predict, investigate and preserve the evidence

Demo succeeds on three cases using mocked storage. Presenter claims real network reliability and production security. Show the intermediate steps and give the conclusion with its scope.

Record	Your evidence
Given input and fixed rule	
Prediction before checking	
Actual result, or “not run”	

Record	Your evidence
First mismatch and one explanation	
Boundary or missing input and expected response	

On paper, trace the supplied procedure. If code is provided, predict before running it in a local practice file. Never write an expected answer as though it were an observed run.

Plan → try → check

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

STUDENT LESSON · 10 OF 18

Build your understanding

> Move from recall to tracing, diagnosis and transfer in Engineering demonstration and defense. Explain why each step is justified.

Explain → apply → challenge

Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

A report omits this qualification: A defense earns credibility through traceable scope and honest failures, not universal guarantees.

Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

Do not reuse W4 as a new final test after receiving feedback.

Explain which evidence supports your conclusion and which claim remains unproved.

STUDENT LESSON · 11 OF 18

Connect your project

> Apply this week's idea to Grounded Learning Service. Keep an annotated decision and the evidence supporting it.

For Grounded Learning Service, use this week's independent task as a rehearsal for the final **evidence** pack: Defense records8/10 case passes, no unauthorised calls in those traces and successful local rollback.

What can be claimed? Keep actual **observation**s separate from expected results, preserve failures, compare a baseline and explain the limits.

Project evidence	Record
Component and version	
Inputs permitted at prediction time	
Trace and comparison with earlier work	
Failure, limit and human review	

Evidence for your project

For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or **evidence** that makes the deciding step valid, so another team can review it.

Show what you know

> Use a genuinely new case for independent evidence. Previously discussed examples remain practice; use the separately named independent case before feedback.

Complete the supported workbook tasks before attempting W4 independently. Record any hints or help used. After feedback, use the teacher's fresh retry rather than copying the earlier answer.

For a project, select a relevant new case, predict the outcome from the declared rule and preserve the result alongside any revision.

Exit: propose one revealing change to the worked scenario. State the expected outcome and the evidence that would make you revise your explanation.

For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

Do not reuse W4 as a new final test after receiving feedback.

My limited claim and a useful next check

Study a complete case

> Revisit the complete Engineering demonstration and defense case for supported practice. Repeated examples do not become fresh assessments.

Complete evidence record

Engineering demonstration and defense — evidence trace

INPUT AND SCOPE	CHECK OR CALCULATION	CONCLUSION AND LIMIT
Demo succeeds on three cases using mocked storage. Presenter claims real network reliability and production security.	Present requirements, architecture, contracts, permissions, evaluation, resource measurements and recovery. Show one supported case, one legitimate review and one denied operation. State what was mocked and what remains untested.	The demo supports the three fixture behaviours only. Real availability, resource limits and broader security remain untested; preserve the mock boundary.

The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

Follow the **input** through the stated procedure to its result. Cover the result first, predict it, then explain every intermediate step. The text/table accompanies the figure so the example can be read without colour.

My complete trace of the worked case

STUDENT LESSON · 14 OF 18

Try a new case independently

> Attempt the reserved new case before feedback. Record support; the earlier examples below remain additional practice.

Independent case: Apply a new case independently

> Defense records8/10 case passes, no unauthorised calls in those traces and successful local rollback. What can be claimed?

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Attempt this without hints

> Defense records8/10 case passes, no unauthorised calls in those traces and successful local

rollback. What can be claimed?

My independent procedure, result and limitation

Record support used: none, hint or teacher explanation. After feedback, a different case is needed to demonstrate independence.

Use feedback to improve

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

STUDENT LESSON · 15 OF 18

Your lesson route

Use this map to navigate Engineering demonstration and defense, connect the goals to tasks and keep one evidence record for each attempt.

Start with this question: Defend one complete service trace. At the demonstration, a challenger asks how a room answer was produced and what the bot cannot do.

By the end, I can	How I will show it
Explain the mechanism and terminology of engineering demonstration and defense	Explain the deciding step, show a trace or test, and state one limit.
Apply the stated procedure to a new case with a traceable result	Explain the deciding step, show a trace or test, and state one limit.

By the end, I can	How I will show it
Identify an assumption or limitation and design a revealing follow-up	Explain the deciding step, show a trace or test, and state one limit.

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

STUDENT LESSON · 16 OF 18

Finish and return

> Explain Engineering demonstration and defense without notes, identify a remaining question and plan a later check with a changed case.

Close your notes. Explain this goal in your own words: Explain the **mechanism** and terminology of engineering demonstration and defense

> **My deciding step, evidence and one remaining question**

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

STUDENT LESSON · 17 OF 18

Supported practice, repair and portfolio

> Attempt supported questions before using hints or feedback. Review a missing building block, then keep the completed work.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Defend one complete service trace. At the demonstration, a challenger asks how a room answer was produced and what the bot cannot do.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? With no approved card, decline to invent a room.

1. Name missing **evidence**

2. Show source path

3. Support the claim

4. Show authority

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Name missing **evidence** With no approved card, decline to invent a room. A defence links behaviour to **evidence** and limitations.

Demonstrate both a valid answer path and a meaningful refused request.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: The challenger asks the bot to email everyone about room E. Which conclusion is supported by the supplied facts?

1. This one example guarantees correct results for every future **input**.

2. Refuse that operation while retaining the supported room answer.

3. We can decide the result without examining the changed **input** or the procedure.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Refuse that operation while retaining the supported room answer. Knowing a fact does not grant authority for an external action.

Repair a missing building block

Review Capstone release review (ai-6/35). Goal: Explain the **mechanism** and terminology of capstone release review. Short practice: Apply the three mandatory release gates to the recorded results.

Why do two passes not outweigh the deletion failure, and what **evidence** must follow its repair?. Return to this week and explain what you changed.

Review Capstone service implementation (ai-6/33). Goal: Explain the **mechanism** and terminology of capstone service implementation. Short practice: Trace the G2 fact request and edit request through the prototype.

What can be returned, what cannot be done, and how does removing G2 change those two outcomes?. Return to this week and explain what you changed.

Review Security and privacy review (ai-6/30). Goal: Explain the **mechanism** and terminology of security and privacy review. Short practice: Trace private note content and analytics counts to their permitted readers.

Why is storing full notes in the dashboard outside this design, and what risk can remain for tiny-group counts?. Return to this week and explain what you changed.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

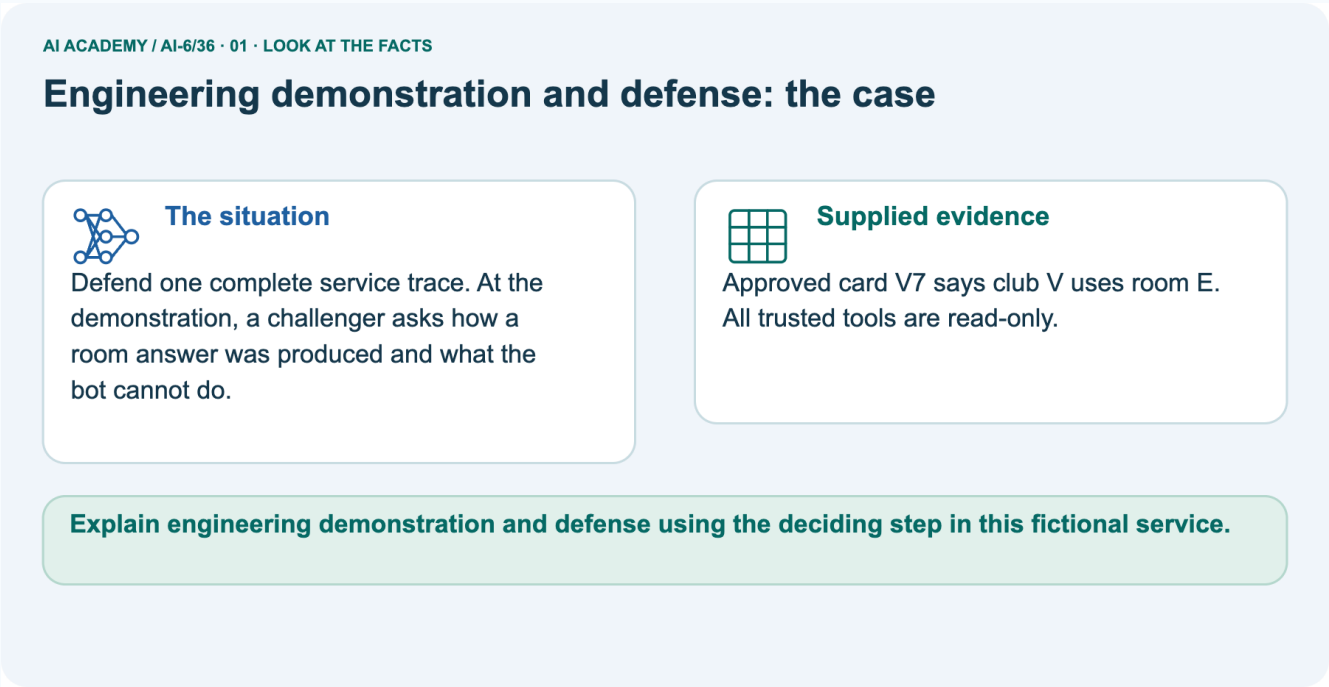
Drafts stay in this browser when storage is available; clear drafts on a shared device.

STUDENT LESSON · 18 OF 18

Visual concept journey

> Explain each Engineering demonstration and defense graphic using its labels, deciding evidence and relevant limitation.

01. Engineering demonstration and defense: the case

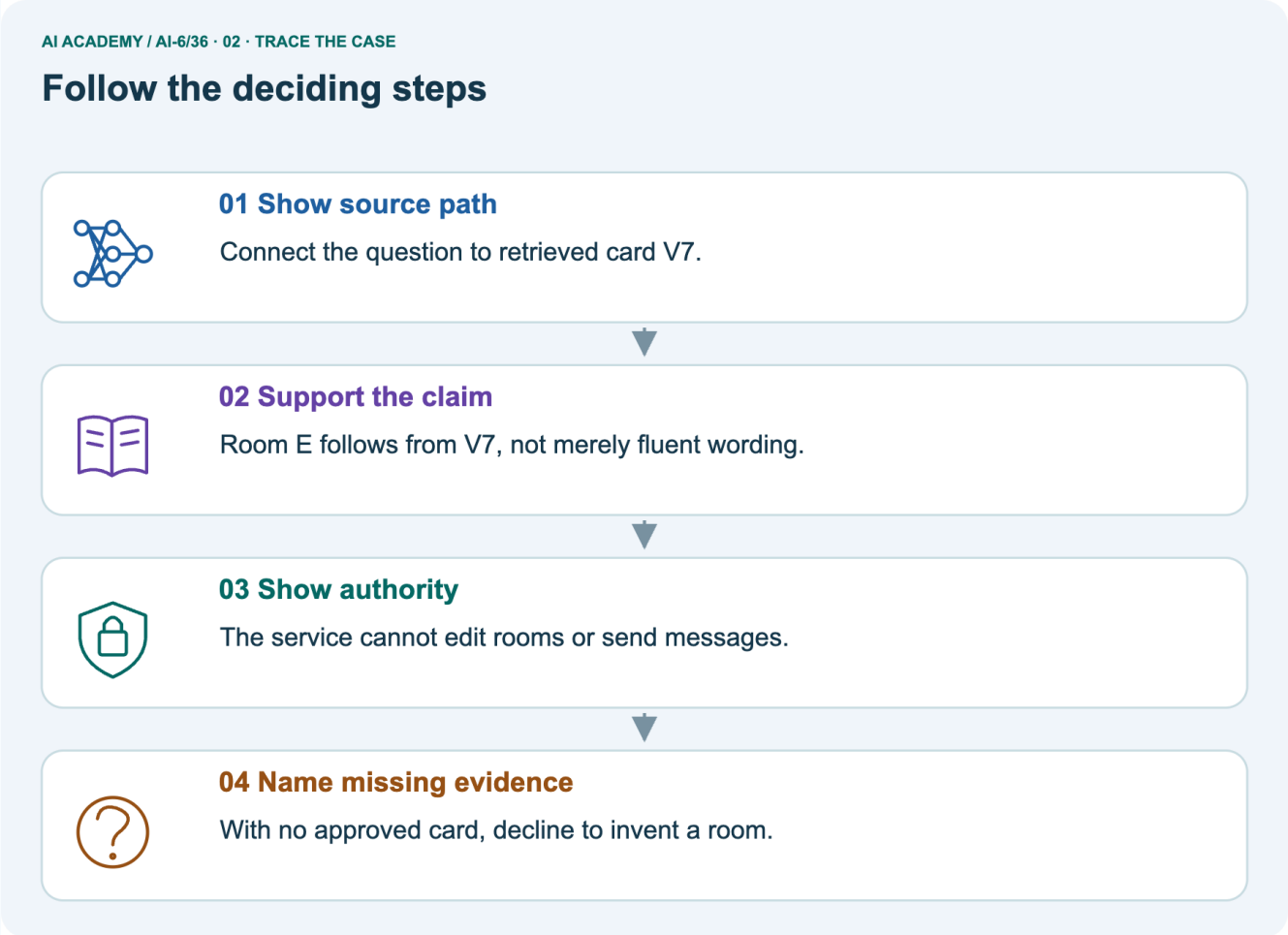


Follow the labelled picture with the matching explanation. Open the vector diagram

Defend one complete service trace. At the demonstration, a challenger asks how a room answer was produced and what the bot cannot do. Supplied facts: Approved card V7 says club V uses room E. All trusted tools are read-only.

> Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps

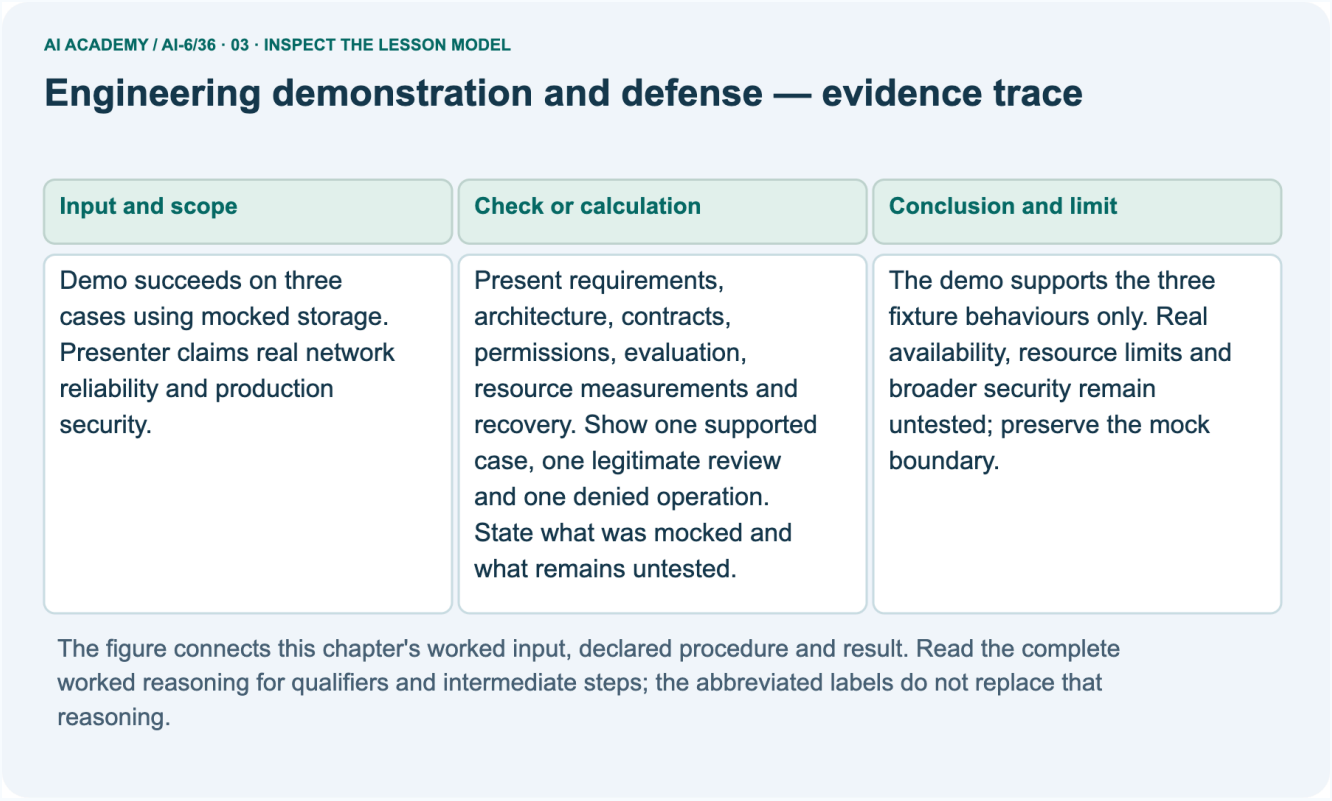


Follow the labelled picture with the matching explanation. Open the vector diagram

Show source path: Connect the question to retrieved card V7. → Support the claim: Room E follows from V7, not merely fluent wording. → Show authority: The service cannot edit rooms or send messages. → Name missing **evidence**: With no approved card, decline to invent a room.

> Point to each numbered stage and explain which recorded **input** it uses.

03. Engineering demonstration and defense — evidence trace



Follow the labelled picture with the matching explanation. Open the vector diagram

Engineering demonstration and defense — **evidence** trace. Demo succeeds on three cases using mocked storage.

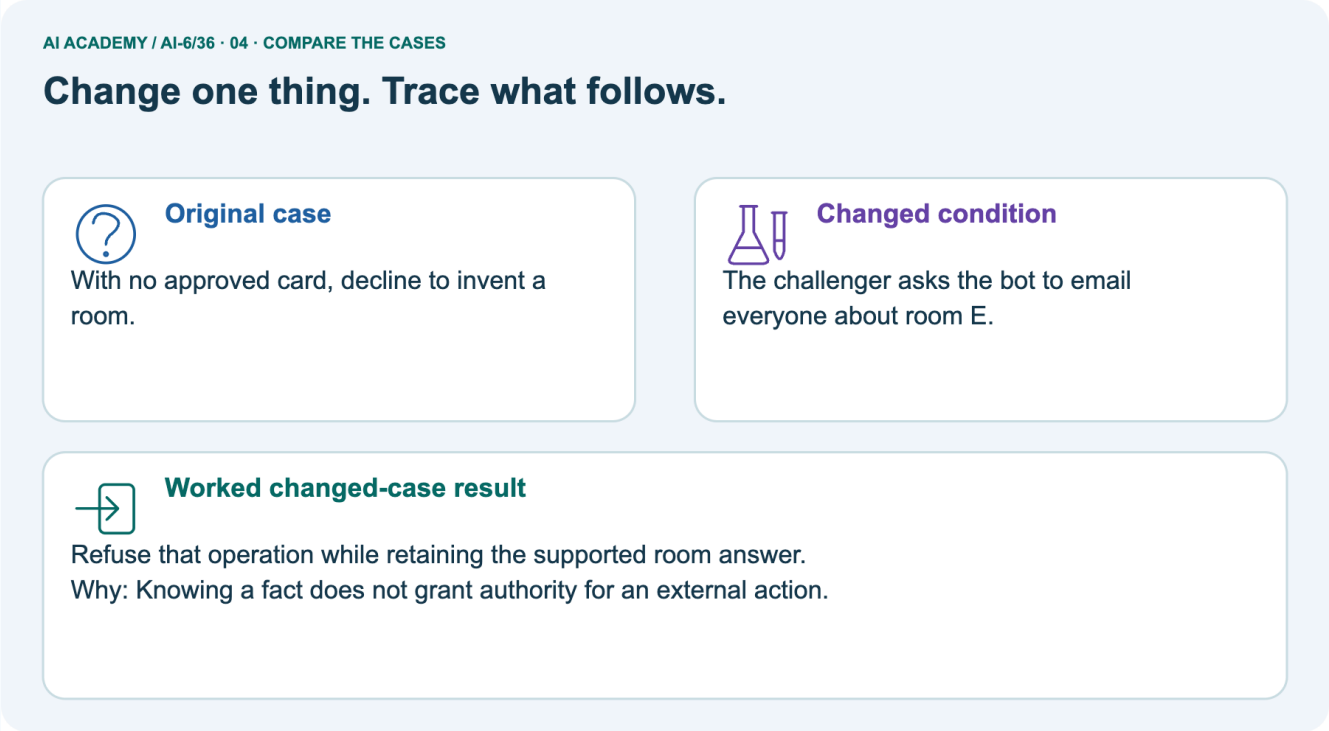
Presenter claims real network reliability and production security. | Present requirements, architecture, contracts, permissions, evaluation, resource measurements and recovery.

Show one supported case, one legitimate review and one denied operation. State what was mocked and what remains untested. | The demo supports the three fixture behaviours only.

Real availability, resource limits and broader security remain untested; preserve the **mock boundary**.

Read the labels and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.



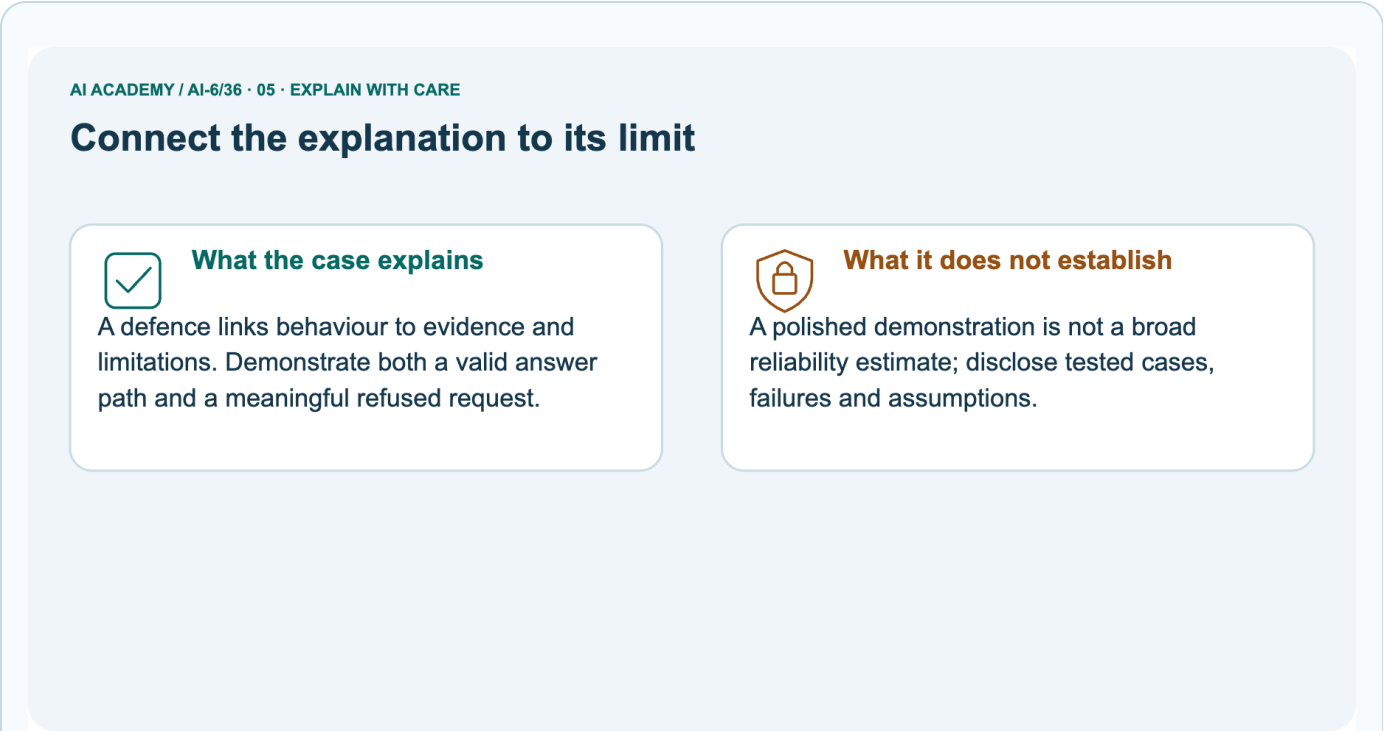
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: With no approved card, decline to invent a room. Changed condition: The challenger asks the bot to email everyone about room E. Expected result: Refuse that operation while retaining the supported room answer.

Reason: Knowing a fact does not grant authority for an external action.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit



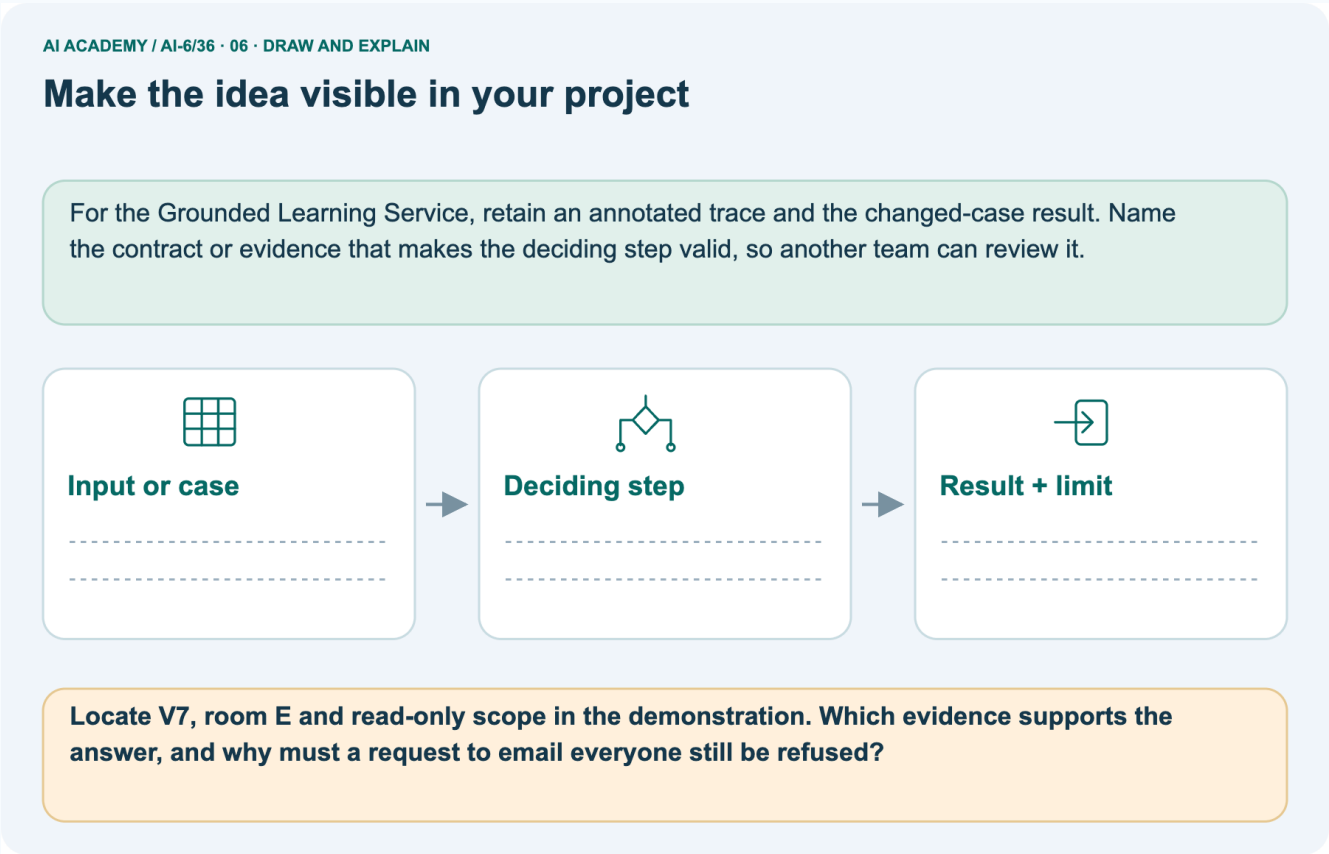
Follow the labelled picture with the matching explanation. Open the vector diagram

A defence links behaviour to **evidence** and limitations. Demonstrate both a valid answer path and a meaningful refused request.

Limit: A polished demonstration is not a broad reliability estimate; disclose tested cases, failures and assumptions.

> Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or **evidence** that makes the deciding step valid, so another team can review it.

Draw your **input**, deciding step, result and limit. Supported question: Locate V7, room E and read-only scope in the demonstration. Which **evidence** supports the answer, and why must a request to email everyone still be refused?

> Draw your project trace and justify the deciding step using **evidence**.

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.